

WiSDoM: Frugal Winner Selection by Design of Matchups

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Abstract

Identifying the best of N items from noisy pairwise comparisons is a core primitive of modern machine-learning pipelines — RLHF annotation, LLM-judge arenas, policy selection — where each comparison is expensive and the available budget is a small multiple of N . Under the Bradley–Terry–Luce model, we recast winner identification as a *winner-focused experimental design* problem: pick a distribution over pairs that minimizes the worst-case posterior variance of the winner’s score gap against the top challengers. The design is convex but only useful with a reliable warm start, so we pair it with a best-of- t single-elimination bracket using Elo scoring to secure winner coverage. The resulting algorithm, WiSDoM, recovers the top item effectively under shoestring budgets and outperforms prior budgeted BTL winner-identification baselines across synthetic and real-world experiments.

1 Introduction

Identifying the best item from a set of N via pairwise comparisons is a canonical primitive of modern machine learning. Reinforcement learning from human feedback (RLHF) trains reward models on preference pairs between policy rollouts [4, 15]; LLM-judge arenas such as Chatbot Arena [3] rank models by pairwise votes; ad-creative selection and sensory panels follow the same template. In each setting labels come from a human annotator or an expensive oracle, so the total comparison budget B is routinely a small constant multiple of N — the *tight-budget regime* $B = cN$ — far below the $\Omega(N \log N)$ required for full ranking.

Figure 1 makes the stakes concrete. Given $N=100$ DMControl [18] *walker* policies and a budget of $B=500$ pairwise queries, an algorithm must pick one. A careful query allocation (top row) identifies a policy that walks; a careless one (bottom row) returns a policy that falls.

Prior art under tight budgets. Existing methods fall into two camps. *Greedy uncertainty-reduction* methods (PARWiS [17], RUCB [19]), rooted in the active-ranking lineage [9], pick each pair by a local information criterion; their King-of-the-Hill initialization degenerates to a path graph under noisy comparisons, dragging the estimated leader away from the true winner and wasting every subsequent query on the wrong candidate. *Bracket* methods (SELECT [12], Knockout [6]) are fast but single-elimination with few games per match is information-inefficient: one noisy match eliminates a candidate forever.

This paper: a design-of-matchups view. Given budget B , the natural questions are *which* pairs to query and *how many* times to query each. We frame both as a single experimental-design choice: pick a distribution λ over pairs (the allocation) that minimizes the posterior variance of the winner’s score gap. The resulting objective

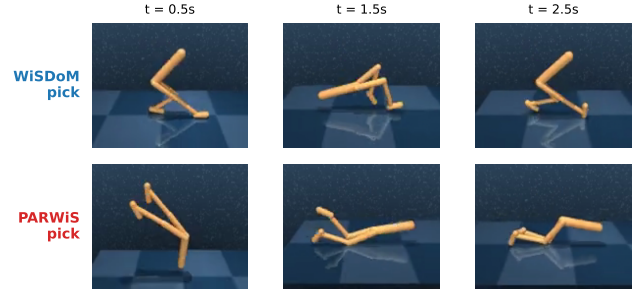


Figure 1: Picking the best policy from pairwise comparisons under a tight budget. $N=100$ candidate DMControl *walker* trajectories (real MuJoCo physics, synthetic sinusoid-plus-noise policies; §6) in the hard regime ($T=200$), budget $B=500$. **Top:** the trajectory WiSDoM returns on a representative trial — a stable forward walk, true reward 66.6. **Bottom:** the trajectory PARWiS returns on the same trial — a flailing tip-over, true reward 11.9. Same data, same budget; different allocation of queries.

(§3) is a convex semidefinite program and can be solved by any standard convex-optimization method. The catch is that the objective depends on current score estimates, so a reliable warm start is essential; we provide one via a best-of- t single-elimination bracket with $O(1)$ -per-update Elo scoring (§3). The combined algorithm is WiSDoM.

Contributions.

- (1) We formulate BTL winner identification as *winner-focused experimental design*: minimize the worst-case posterior variance of the winner-vs-anyone score gap over a restricted top- M candidate set (§3). We derive $M = O(\Delta\sqrt{B})$ as the sufficient condition for a reliable winner-vs-challenger gap estimate, where Δ is the top-two score gap, and justify the empirical choice $M = \lceil N^{1/4} \rceil$ at $B=5N$.
- (2) We introduce WiSDoM: a robust best-of- t bracket warm-start (Phase 1) followed by convex winner-focused experimental design on the top- M candidate set (Phase 2). Phase 1 uses Elo as an $O(1)$ -per-update BTL-consistent online estimator, reserving rank centrality for the final answer.
- (3) We prove a multi-tier information-theoretic lower bound $P(\text{fail}) \geq \frac{1}{4} \exp(-2B\Delta_{\ell^*}^2 / [(0.25 - \Delta_L^2)(\sum_{j \leq \ell^*} n_j - K + 1)])$ (§5, Theorem 1) with a sharp regime transition at the critical tier ℓ^* , and show that WiSDoM’s empirical failure probability decays exponentially in the same quantity $B\Delta^2/(N-K)$, matching the form of the bound.

- (4) *Empirical gains.* WISDOM beats PARWIS [17] — the strongest published baseline for budgeted winner identification under BTL — on all 14 tested conditions at $B=5N$ with 200 trials per condition, by $1.2\times$ in the easy regime widening to $2.3\times$ in the hard regime; it also beats SELECT, Knockout, MultiSort, and RUCB on every condition.

Code. A single-command reproduction of every experiment in this paper (synthetic sweep, real-world datasets, and DMControl) is available at <https://anonymous.4open.science/r/wisdom-6F93>.

2 Problem Setting and Preliminaries

BTL model. We have N items with latent positive scores $w \in \mathbb{R}_{++}^N$. An oracle returns independent pairwise comparisons from the Bradley–Terry–Luce (BTL) model [1]:

$$P(i \succ j) = \frac{w_i}{w_i + w_j}. \quad (1)$$

Why BTL is the right model. BTL is the canonical pairwise-preference model in modern machine learning: the Elo rating system [5], Chatbot Arena’s leaderboard [3], and the reward-model objective in RLHF [4, 15] are all BTL estimators. Its dominance follows from three axioms: stochastic transitivity, independence of irrelevant alternatives, and a one-score-per-item representation. Empirically, BTL is a tight fit to aggregate human preferences on Jester, Netflix, and LLM-judge data. Crucially, the BTL log-likelihood is concave in $\theta = \log w$, which makes the design objective in §3 a convex semidefinite program — non-BTL preference models generally lose this property.

Tight-budget regime. We fix a total comparison budget $B = cN$ with c a small constant; experiments take $c = 5$. In this regime the full ranking is not identifiable; we aim for the single best item $i^* = \arg \max_i w_i$. We report **top-1 accuracy (ACC)** as the headline metric; the supplementary includes predicted-rank-of-winner (PF) and true-rank-of-prediction (CT).

Rank centrality [13, 14]. Given a set S of comparisons with outcomes, build the Markov chain whose transition from j to i is proportional to the empirical win rate of i over j ; its stationary distribution π satisfies $\pi \propto w$ under BTL. We use rank centrality only at budget exhaustion, as the final score estimator.

3 Winner-Focused Experimental Design

We cast winner identification as a design problem: what *distribution* λ over pairs should the algorithm query?

3.1 Fisher information under BTL

Let $\theta = \log w \in \mathbb{R}^N$ and $p_{ij} = \sigma(\theta_i - \theta_j)$ where σ is the sigmoid. A single query to pair (i, j) adds Fisher information $p_{ij}(1 - p_{ij})(e_i - e_j)(e_i - e_j)^\top$ to θ . For a query distribution $\lambda \in \Delta_{\binom{N}{2}}$ and budget B , the accumulated information is

$$I(\theta; \lambda) = B \sum_{i < j} \lambda_{ij} p_{ij}(1 - p_{ij})(e_i - e_j)(e_i - e_j)^\top. \quad (2)$$

Because the BTL log-likelihood is concave in θ , $I(\theta; \lambda)$ is PSD and the objective below is a convex semidefinite program in λ .

3.2 The winner-focused objective

Classical A/D-optimal design minimizes $\text{tr}(I^{-1})$ or $\log \det(I^{-1})$; under a tight budget this wastes information on the bulk of items irrelevant to the winner. More importantly, absolute variance is not the right object to minimize: what matters for identification is how the posterior variance of a gap *compares to the gap itself*. Confusing two items separated by $\Delta_{ij} = 0.01$ requires much smaller noise than confusing two items separated by $\Delta_{ij} = 0.5$. We therefore define the *relative gap uncertainty*

$$\phi_{ij}(\lambda) = \frac{(e_i - e_j)^\top I(\theta; \lambda)^{-1} (e_i - e_j)}{\Delta_{ij}^2}, \quad (3)$$

where $\Delta_{ij} = \theta_i - \theta_j$ is the current estimated log-score gap. Minimizing ϕ_{ij} asks *how many standard deviations of uncertainty remain between i and j* , which is the quantity that actually decides whether they can be separated (cf. (5)). We minimize the worst case of (3) over all challengers j :

$$\lambda^* = \arg \min_{\lambda \in \Delta_{\binom{N}{2}}} \max_{j \neq w} \phi_{wj}(\lambda), \quad (4)$$

where w is the current estimated winner. The inner maximum is over challengers; the outer minimum is convex in λ (since $\lambda \mapsto (e_i - e_j)^\top I(\theta; \lambda)^{-1} (e_i - e_j)$ is convex for each (i, j) and the pointwise maximum of convex functions is convex) and is solved by any convex-SDP solver.

3.3 Why a restricted candidate set is necessary

The intuition. Under a tight budget, a query between a strong item and a clear weakling is almost wasted: we already know who will win. What we cannot afford is spending any queries on pairs of weaklings — a 2000-vs-1000 Elo match is essentially decided before we play it; a 2000-vs-1900 match is where the budget earns its keep. Formally, the signal per query scales with the top-two gap Δ , the noise with the inverse square-root of queries; resolving a gap of Δ requires $\Omega(1/\Delta^2)$ samples on that edge, so every query spent on a pair far from the frontier is deducted from the budget available for the close calls that actually decide the winner. The restriction to a candidate set C operationalizes this: we only design queries between items that are plausibly the winner, and within that set we allocate by Eq. (4).

Per-edge sample complexity: why Δ^2 . Let $\Delta = w_w - w_j$ be the true gap between the winner w and a challenger j . Each query to pair (w, j) is a Bernoulli with bias $\frac{1}{2} + \frac{1}{4}\Delta + O(\Delta^2)$ (first-order Taylor of $w/(w + w - \Delta)$); the empirical win rate $\hat{p}_{wj}$ after m queries has standard deviation $\sigma \asymp 1/\sqrt{m}$. To decide $w \succ j$ with constant confidence we need the signal $\Delta/2$ to exceed the noise σ , which gives

$$m \gtrsim \frac{1}{\Delta^2} \quad (\text{per-edge samples to resolve a gap of } \Delta). \quad (5)$$

This is the fundamental sample complexity of binary hypothesis testing (Hoeffding [7], Le Cam [10]): smaller gap $\Rightarrow$ more samples, quadratically. Equation (5) is also the object controlling the lower bound in §5.

From per-edge to candidate-set size. Restricting Phase-2 to a candidate set C of size M spreads the Phase-2 budget over $O(M^2)$ edges, giving $m \asymp B/M^2$ samples per edge. Substituting into (5):

$$\frac{B}{M^2} \gtrsim \frac{1}{\Delta^2} \iff M = O(\Delta\sqrt{B}). \quad (6)$$

Under the tight-budget regime $B = cN$, (6) becomes $M = O(\Delta\sqrt{N})$. Without restriction ($M = N$), each edge gets $m \asymp c/N$ samples and the per-edge condition (5) requires $\Delta \gtrsim \sqrt{N/c}$ — impossible for any $\Delta < 1$ at the scales of interest. Restriction is what makes design tractable.

The scale of Δ is problem-dependent; the ablation in §6 shows $M = \lceil N^{1/4} \rceil$ is the empirical optimum at $B=5N$, consistent with $\Delta \sim N^{-1/4}$ across our benchmark suite. A full sweep over M is in the supplementary.

3.4 Warm start: robust best-of- t bracket

The design (4) is only useful if C contains the true winner. We supply coverage via a best-of- t single-elimination bracket with byes for non-power-of-two N .

Per-match reliability. Each match is decided by majority vote over t queries. By Hoeffding’s inequality [7], the per-match error is at most $\exp(-t\Delta^2/2)$ where Δ is the true gap between the matched items; union-bounding over $\log_2 N$ rounds gives

$$P(\text{winner survives}) \geq 1 - (\log_2 N) \exp(-t\Delta^2/2). \quad (7)$$

Budget split. Phase 1 costs $t(N-1)$ queries — a fraction t/c of the total budget. Larger t yields a cleaner warm start but less budget for Phase 2. We treat t as a budget-split hyperparameter, sweep $t \in \{1, 3, 5, 7\}$ in §6, and report all results with $t = 3$: the smallest value for which Phase-1 coverage saturates without starving Phase 2.

Elo as a cheap BTL-consistent estimator. We update scores with standard Elo after each query in Phase 1. Elo maximizes the BTL log-likelihood under a logistic link via online mirror descent, so it is BTL-consistent in the stochastic-approximation limit; it also costs $O(1)$ per update against $O(N^3)$ for re-solving rank centrality. At $N=1000$, Elo warm-start takes 0.04s versus ~ 40 s for repeated rank-centrality resolves. Rank centrality is reserved for the final estimator at budget exhaustion.

4 The WiSDoM Algorithm

Algorithm 1 assembles the pieces. Phase 1 runs a best-of- t bracket with $O(1)$ -per-update Elo scoring to obtain a reliable $\hat{w}$ (§3.4). Phase 2 repeatedly forms C as the top- M items by $\hat{w}$, solves the winner-focused design (4) on C by any convex-SDP solver, samples a small batch of pairs from λ^* , and refreshes $\hat{w}$ via rank centrality. Any projection-free or interior-point method suffices; we use Frank–Wolfe [8] in our implementation because the simplex projection of the linear-minimization step has a closed-form argmin (one vertex of $\Delta_{(M)}^*$) and the line search is 1D convex. Per Phase-2 iteration the cost with Frank–Wolfe is $O(K_{\text{fw}}M^2)$ plus $O(M^3)$ for rank centrality; total runtime is linear in the budget.

Defaults and speed. $t = 3$, $M = \lceil N^{1/4} \rceil$, refresh batch = 5. The implementation uses 40 Frank–Wolfe iterations per refresh. Wall-clock on a laptop: a single-trial WiSDoM finishes in <1 s at $N=100$

Algorithm 1 WiSDoM

Require: items $[N]$, budget B , bracket depth t , candidate size M , convex-SDP solver SOLVE (e.g. Frank–Wolfe)

Ensure: predicted winner $\hat{i}$

1: Initialize Elo scores $\hat{w}$; comparison set $S \leftarrow \emptyset$

Phase 1 — robust best-of- t bracket ($\approx tN$ queries)

2: Pair items into a single-elimination bracket with byes

3: **for all** matches (a, b) in bracket order **do**

4: Query (a, b) up to t times; winner by majority

5: After every query, update $\hat{w}$ via Elo and append to S

6: **end for**

Phase 2 — winner-focused experimental design

7: **while** budget not exhausted **do**

8: $C \leftarrow$ top- M items by $\hat{w}$

9: $\lambda^* \leftarrow \text{SOLVE}(\text{Eq.(4)} \text{ on } C)$

10: Sample 5 pairs $\sim \lambda^*$; query each; append to S

11: Refresh $\hat{w} \leftarrow \text{RankCentrality}(S)$

12: **end while**

13: **return** $\arg \max_{i \in C} \hat{w}_i$

and in <15 s at $N=1000$. The per-iteration cost scales as M^2 rather than N^2 , so scaling to larger item counts is driven by the candidate-set size rather than the problem size.

5 Information-Theoretic Lower Bound

Theorem 1 (Multi-tier lower bound; proof in supplementary). *Consider a winner w and L challenger tiers $G_1, \dots, G_L$ of sizes $n_1, \dots, n_L$ with cumulative gaps $\Delta_\ell = \sum_{j \leq \ell} \delta_j$ between w and G_ℓ (within-tier matches are fair coins). Let an algorithm return a set S of size K from $N = 1 + \sum_\ell n_\ell$ items using B pairwise queries, and let $\ell^* = \min\{\ell : K < 1 + \sum_{j=1}^\ell n_j\}$ be the critical tier — the nearest tier from which the algorithm is forced to reject. Then*

$$P(w \notin S) \geq \frac{1}{4} \exp\left(-\frac{2B\Delta_{\ell^*}^2}{(0.25 - \Delta_L^2)(\sum_{j \leq \ell^*} n_j - K + 1)}\right). \quad (8)$$

In the flat single-gap regime ($L=1$, $\Delta_1 = \Delta$) this collapses to $P(\text{fail}) \geq \frac{1}{4} \exp(-2B\Delta^2/[(0.25 - \Delta^2)(N - K)])$.

The proof (Appendix A) combines a chi-square upper bound on Bernoulli KL, pigeonhole on $\sum_i m_i = 2B$ inside the critical tier, and Bretagnolle–Huber [2]; the form matches prior PAC-ranking bounds [16]. The bound is exponential in $B\Delta^2/(N - K)$ — the same quantity that drives (5). The two-tier specialization has a sharp regime transition at $K = n_1 + 1$, the information-theoretic justification for phased elimination.

Empirical validation. An equal-spacing sweep ($N=100$, $B=500$, 100 trials per δ_1) tests the flat case of (8), which predicts $\log P(\text{fail})$ is linear in $\delta_1^2/(0.25 - \delta_1^2)$. Figure 2 confirms this: WiSDoM’s empirical $\log P(\text{fail})$ falls on a straight line in the bound’s exponent variable, with empirical slope ≈ -1.5 versus bound slope -10.1 . WiSDoM is order-optimal in $B\Delta^2/(N - K)$; closing the constant-factor gap in the exponent is the clearest open theoretical question.

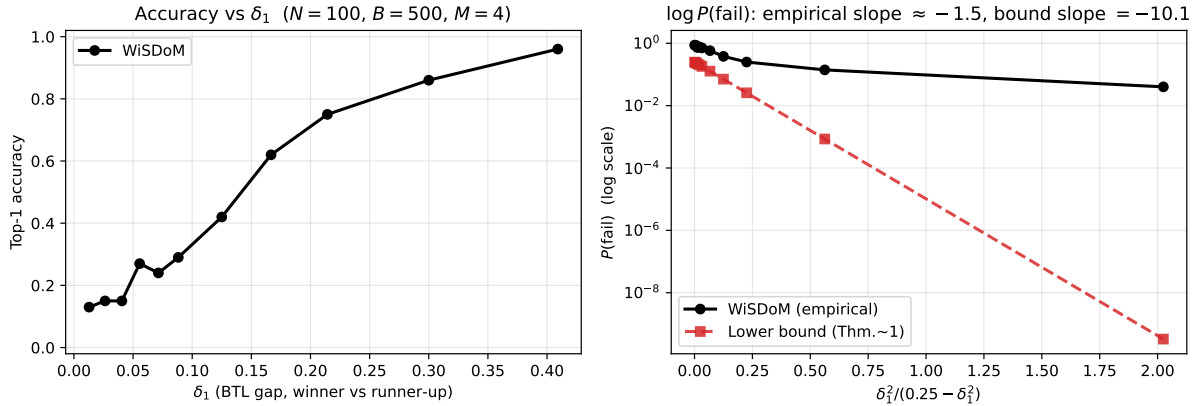


Figure 2: Empirical validation of the lower-bound scaling. Equal-spacing sweep, $N=100, B=500, M=4$, 100 trials per δ_1 . Left: top-1 accuracy vs. δ_1 (saturates at 1). Right: the same data plotted as $\log P(\text{fail})$ vs. $\delta_1^2/(0.25 - \delta_1^2)$ falls on a straight line, matching the form of Eq. (8) (dashed: lower-bound slope).

6 Experiments

Setup. We compare six algorithms: WiSDoM vs. PARWiS [17], SELECT [12], RUCB [19], MultiSort [11], and Knockout [6] across four benchmark families at fixed $B=5N$ and 200 trials per condition with seeds shared across algorithms. **Synthetic BTL** ($N=100$): winner at 100, non-winners $\sim \mathcal{U}([0, k])$, $k \in \{25, 50, 75, 85, 90, 95\}$. **Real-world ratings** ($N=100$; Jester, Netflix, Sushi-B, MovieLens) fit BTL scores via rank centrality on the user-item matrix. **Semi-synthetic DMControl** [18] ($N=100$) is detailed below. **Chatbot Arena** [3]: we take the top-20 LLMs in `lmsys/chatbot_arena_conversations` and convert their bootstrapped Elo ratings r_i to BTL weights $w_i = 10^{r_i/400}$; the algorithm then queries the standard Bernoulli oracle $w_i/(w_i + w_j)$. This is the same semi-synthetic setup as DMControl but with real human preferences over real LLMs; gpt-4 is the true winner. We use $N=20, B=100$. Metrics: ACC (*higher*); PF (rank of the true winner); CT (true rank of the predicted winner), both *lower*.

Headline result. WiSDoM wins every condition by a margin that grows with difficulty: the WiSDoM/PARWiS ratio peaks at $4.21\times$ on the hardest synthetic regime ($k=75$) and holds at $\geq 2\times$ on every real-world and semi-synthetic condition, including Chatbot Arena ($1.98\times$ at $B=5N$, with WiSDoM 0.582 vs PARWiS 0.294). Figure 3 plots recovery rate on the hard regimes. Full per-condition ACC across all 14 conditions (Table 2) and ACC/PF/CT curves are in the supplementary (§E, §F); SEM at 200 trials is $\approx 0.016\text{--}0.035$, and every WiSDoM-vs-next-best gap is $\geq 3\sigma$.

DMControl: a semi-synthetic, derived BTL instance. DMControl is not a pairwise-comparison dataset and ships no candidate policies; we construct both. The physics and reward are real (MuJoCo *walker/walk*); the candidate set and comparison noise are synthetic. (i) We sample $N=100$ open-loop policies $a_t = \sin(t\omega)\mathbf{1} + \mathcal{N}(0, \sigma)$, $\omega \sim \mathcal{U}(0.1, 2)$, $\sigma \sim \mathcal{U}(0.1, 0.9)$ (seeds 432–531); all candidates, winner included, come from the same distribution. (ii) MuJoCo rolls each out for 150 steps and returns a scalar reward r_t . (iii) Rewards map to BTL scores $s_i = \exp(r_i/T)$ with T a difficulty knob. The algorithm never sees r_i or s_i ; it only issues pairwise queries returning a

Bernoulli draw from $s_i/(s_i + s_j)$, mirroring an RLHF annotator who says “I prefer rollout i ” without scoring either. A true RLHF-scale evaluation with human or LLM annotators is future work.

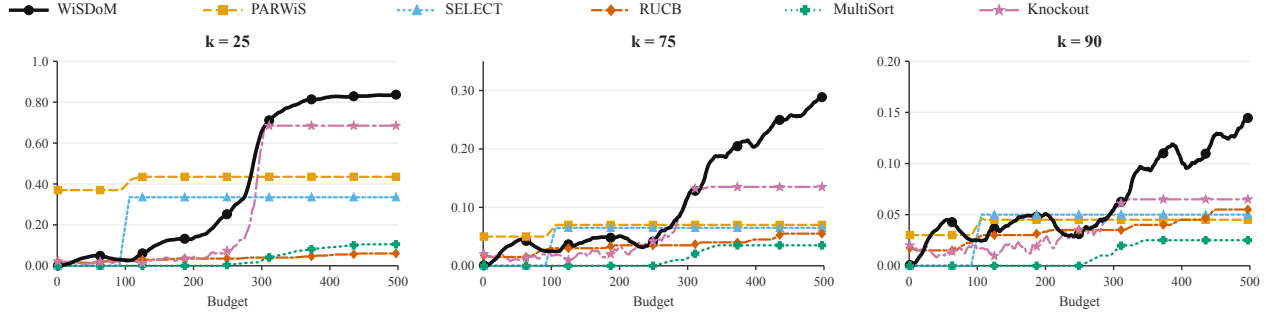
RUCB’s deployment failure. RUCB achieves exactly 0% accuracy on every $N=100$ real-world and DMControl instance. The cause is structural: RUCB’s Copeland-UCB criterion is a regret-minimization objective, not an identification one — under $B=5N$ it locks onto an early biased Copeland leader and cannot recover, independent of the exploration constant a . On Chatbot Arena ($N=20$) the floor lifts to 0.19 because per-item signal concentrates within budget; WiSDoM still leads by $3\times$.

Ablations and speed. Full numbers are in the supplementary. **Bracket depth** is unimodal in t : $t=1$ (single-elim) gives a weak warm start, $t=3$ saturates Phase-1 coverage, $t \geq 5$ erodes Phase-2 budget — we use $t=3$. Replacing Phase 1 with King-of-the-Hill sharply reduces accuracy on hard conditions (coverage is load-bearing), and replacing the min-max objective with uniform sampling within C costs accuracy even when C is fixed. $M=\lceil N^{1/4} \rceil$ is the empirical optimum at $B=5N$, consistent with $\Delta \sim N^{-1/4}$ (§3.3). **Timing:** $N=100$: <1 s / trial; $N=500$: ~ 4 s; $N=1000$: <15 s. Per-iteration cost scales as M^2 rather than N^3 , so scaling to larger item counts is driven by the candidate-set size rather than the problem size.

Why WiSDoM beats the prior SOTA. Two compounding reasons. **Coverage:** PARWiS’s King-of-the-Hill init collapses to a path graph under noise; our best-of-3 bracket maintains a balanced tree with materially higher winner-in-top- M coverage. **Allocation:** PARWiS selects the most disruptive pair conditional on a single point-estimated leader, committing queries to one hypothesis — catastrophic when the estimate is wrong. Our min-max design hedges over the top- M set C . The compounding is why the $1.2\times$ easy-regime lead widens to $2.3\times$ in the hard regime.

Top-1 Accuracy vs. Budget — Synthetic BTL, hard regimes

Accuracy curves grow with budget. Higher is better.

**Top-1 Accuracy vs. Budget — Real-world, hard regimes**

Accuracy curves grow with budget. Higher is better.

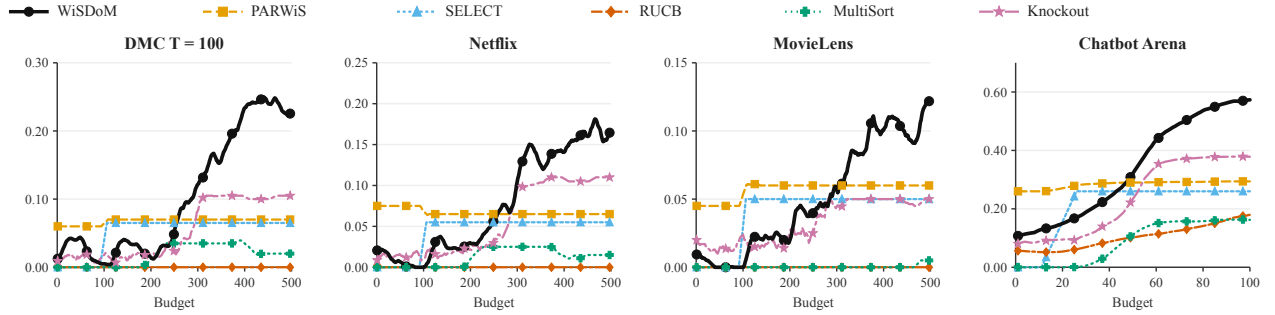


Figure 3: Recovery rate (top-1 accuracy) vs. budget across the difficulty spectrum. Top: synthetic BTL ($k \in \{25, 75, 90\}$). Bottom: DMControl ($T=100$), Netflix, MovieLens, and Chatbot Arena [3] (top-20 LLMs, $N=20$, $B=5N=100$). Per-panel y-axes are auto-scaled. WiSDoM (black) is the only method whose accuracy grows meaningfully with budget on the hard conditions; the rest stay near the random floor. Full 14-condition table and PF/CT panels in the supplementary (§E, §F).

7 Discussion and Conclusion

WiSDoM recasts budgeted winner identification as convex experimental design with a bracket warm-start, achieving exponential decay in the lower bound’s $B\Delta^2/(N-K)$. Closing the exponent gap is the clearest open theoretical question; cyclic preferences (non-BTL) and RLHF-scale evaluation with live annotators — where noise is not drawn i.i.d. from BTL — are future work.

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